

To: Gentari Hydrogen Sdn Bhd



Confidential

IHI business introduction of Ammonia Value Chain

IHI

27 August, 2026

IHI Corporation

Ammonia Value Chain Project Department,
Business Development Headquarters

IHI Power Systems Co., Ltd

IHI Company Overview

Confidential

IHI

IHI is a global company with 23 overseas bases and presence in 22 countries



Year of establishment

1853

(more than 170 years)



Revenue (Consolidated)

Approx. **1,627** billion JPY
(370 billion THB @4.4)



27,990
employees
(consolidated)



Domestic base

■ Core Business

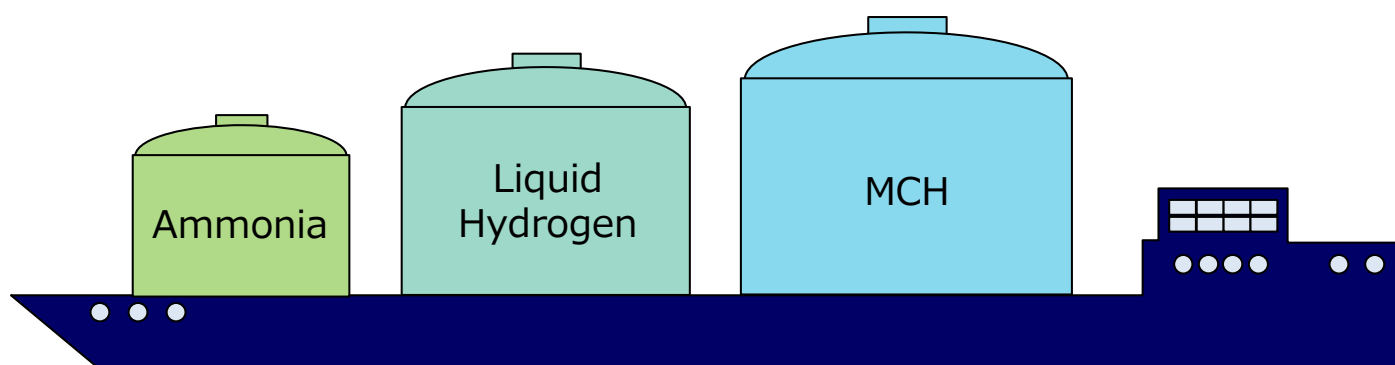
4

Areas of expertise



Why Ammonia?

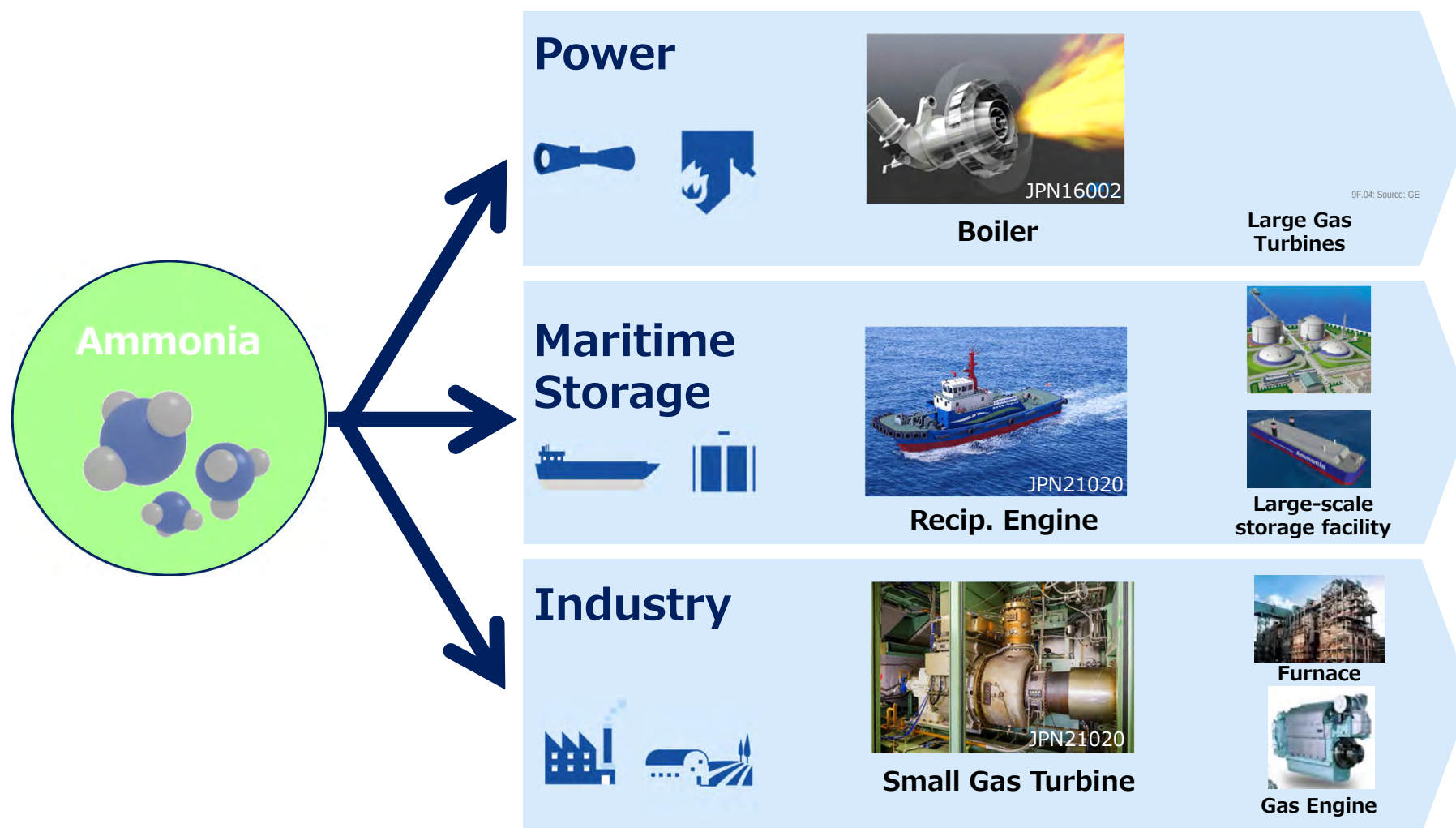
- **Maximum Volumetric Hydrogen Density** compared to other hydrogen carriers which can minimize its transportation cost and size of infrastructure.
- **Easy to Liquefy** ($-33^{\circ}\text{C}@1\text{ atm}$ / $20^{\circ}\text{C}@8.46\text{ atm}$)
- **Well-developed technology** for synthesis and distribution networks in industries such as fertilizer, chemical raw material, etc.
- **Direct use** as Carbon-neutral fuel without CO_2 emissions



	Ammonia	Liquid H2	MCH
Tank Capacity Ratio	1.0	1.5x	2.2x
Tank Material	C/S (SLA325A)	SUS	C/S
Boiling Point @1atm [deg-C]	-33	-253	101
Storage Period	Long-term	7 days	Long-term

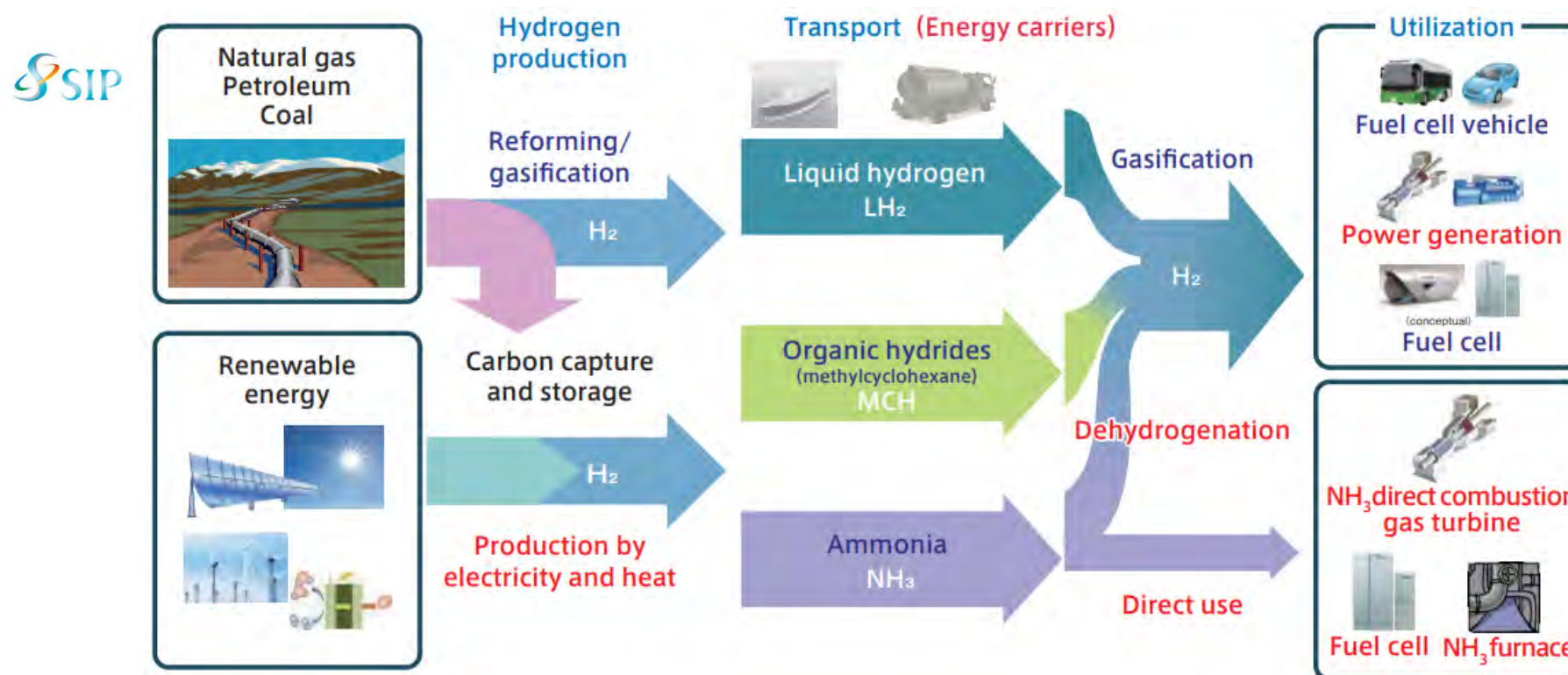
Why Ammonia?

- Ammonia is a decarbonized fuel and is also promising as a hydrogen carrier.
- IHI is promoting technology development for direct use of ammonia in the fields of power, maritime and industry.



Advantages of Ammonia as a Hydrogen Carrier

- **Maximum Volumetric Hydrogen Density** compared to other hydrogen carriers which can minimize its transportation cost and size of infrastructure. (Approx. 11% : 150% of Liquid Hydrogen)
- **Easy to Liquefy** ($-33^{\circ}\text{C}@1\text{atm}$ / $20^{\circ}\text{C}@8.46\text{ atm}$)
- **Well-developed technology** for synthesis and distribution networks for Fertilizer, Chemical raw-material etc. industries
- **Direct use** as Carbon Neutral Fuel without CO₂ emissions



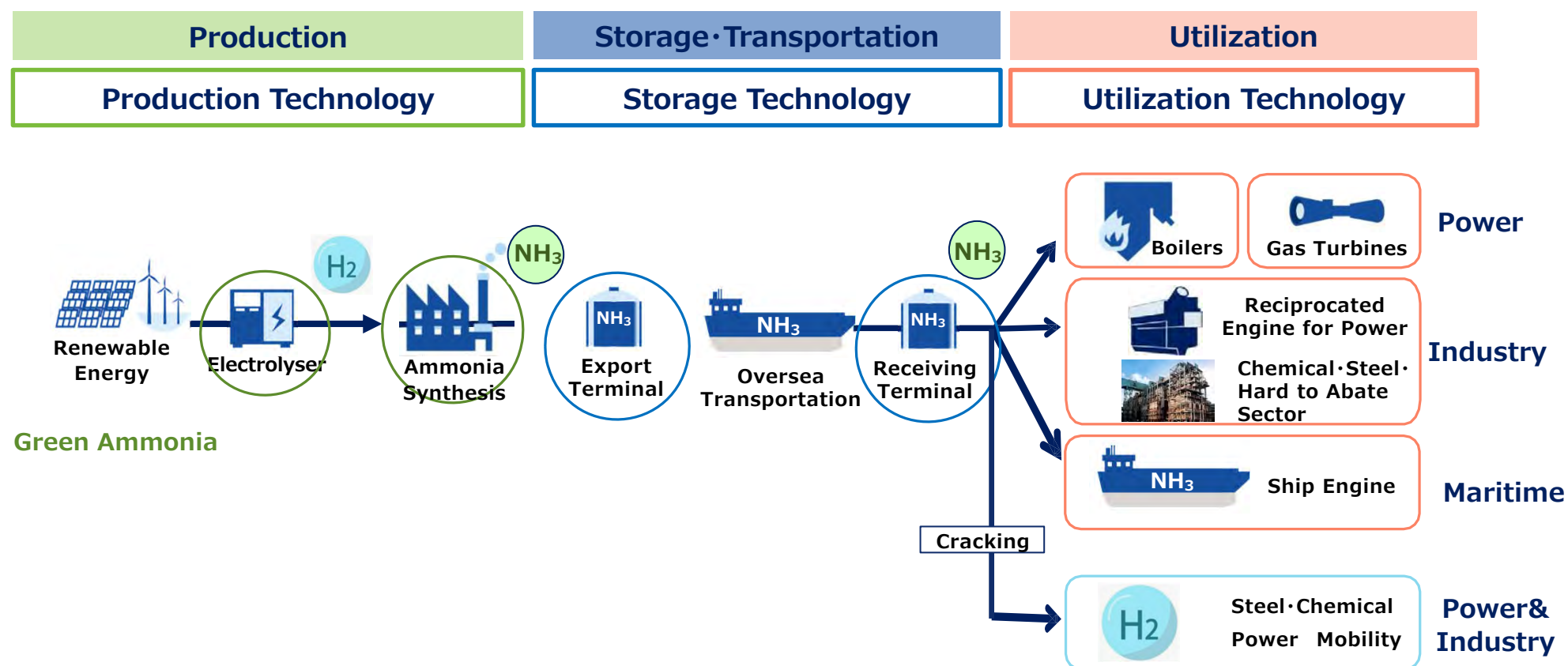
Source ; SIP(Cross-ministerial Strategic Innovation Promotion Program) Energy Carriers

IHI Mission

IHI's mission is to provide **advanced ammonia utilization technologies** and a **reasonable price of ammonia** to partners aiming to achieve carbon neutrality.

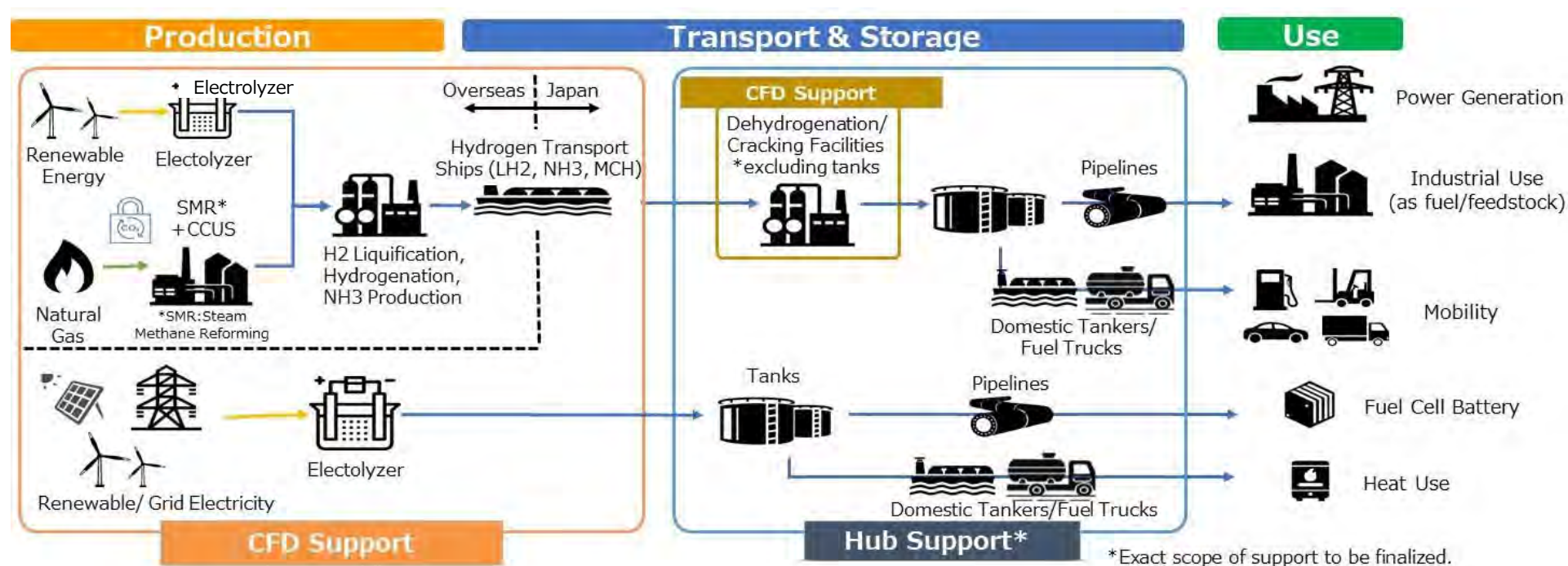
IHI's wide range of **development activities to build an ammonia value chain** include -

- ① Ammonia Production; NH₃ Direct Synthesis
- ② Transportation and Storage; Large Capacity Tanks
- ③ Utilization; Boilers, GTs, Recip. Engines, Package Boilers, Industrial Furnaces
- ④ Standardization for Safe Use



Hydrogen Hub Development Program (JAPAN Case)

- The Hydrogen Hub Development Program supports the establishment of infrastructure which leads to large-scale expansion of the use of low-carbon hydrogen and its derivatives and widely benefits a variety of companies, with an aim to stimulate demand creation and the efficient buildout of hydrogen supply chains.
- The Program will subsidize a portion of the CAPEX for developing “facilities necessary to transport low-carbon hydrogen from the receiving terminal to the point of actual use by consumers and used by multiple companies (e.g. shared pipelines and tanks)”



Green Ammonia Production and Sales Project in India

IHI and ACME Signed Term Sheet to Supply Green Ammonia from India to Japan

- Jan 23, 2024 - Press Release

IHI and ACME Group, a leading renewable energy company in India, today signed a term sheet for the supply of green ammonia from Odisha, India to Japan. The term sheet was signed by Hiroshi Ide, President and CEO of IHI, Shri Manoj Upadhyay, Founder and Chairman of Acme Group in the presence of Union Minister for Power and New & Renewable Energy Shri RK Singh, New & Renewable Energy Secretary Shri Bhupinder Singh Bhalla, and Ambassador of Japan to India Hiroshi Suzuki. Both companies intend to develop the term sheet into a legally binding document within this year.

Under this term sheet, **IHI considers off-taking up to 400,000 tons** of green ammonia **from 2028**, which will be supplied to various industrial customers located mainly in Japan. Both companies are in discussion to partner across the value chain, starting from production to logistics, supply to customers and creation of the green ammonia market for use in a range of applications in power generation and industrial uses in Japan to reduce overall emissions across the value chain.

【URL】 https://www.ihico.jp/en/all_news/2023/resources_energy_environment/1200584_3523.html



Green Ammonia Production and Sales Project in India

Large-Scale Low-Carbon Ammonia Supply Chain Plan Certified under Japan's "Support focusing on the price gap" scheme

- June 30, 2026 -

Press Release

IHI Corporation have announced that IHI's plan to utilize low-carbon ammonia produced in India has been certified under the "Support focusing on the price gap" scheme, in accordance with the Hydrogen Society Promotion Act, by the Minister of Economy, Trade and Industry and the Minister of Land, Infrastructure, Transport and Tourism.

In this supply chain development, IHI will offtake the ammonia produced by a low-carbon ammonia manufacturing project being jointly developed with ACME Group in Odisha, located in eastern India, and supply it to the seven companies in Japan.

【URL】 https://www.ihi.co.jp/en/all_news/2026/resources_energy_environment/1202098_13814.html

Ammonia Supply Hub in Japan (Tomakomai, Hokkaido) IHI

Ammonia Supply Hub Project Based in the Tomakomai Area of Hokkaido Acquires Certification under Japan's Hub Development Support Program

-March 27, 2026-

Press Release

IHI have announced the receipt of approval from the Minister of Economy, Trade and Industry (METI) and the Minister of Land, Infrastructure, Transport and Tourism (MLIT) for certification under the Hub Development Support Program within the Hydrogen Society Promotion Act.

The Hub Development Support Program, implemented by METI's Agency for Natural Resources and Energy, aims to accelerate the supply and utilization of low-carbon hydrogen and derivatives by providing support for the development of infrastructure necessary for transportation and storage.

【URL】 https://www.ihl.co.jp/en/all_news/2025/resources_energy_environment/1201933_13737.html

Jera

IHI
Realize your dreams



Survey on the current status for establish of Ammonia Supply Chain / Ammonia Value Chain

Confidential

IHI

1. Survey on Ammonia Fuel Procurement

- 1) Survey on Global Ammonia Production Status
- 2) Trend Survey on Future Ammonia Procurement

2. Survey on Maritime Transport of Ammonia Fuel

3. Survey on Ammonia Fuel Storage

- 1) Survey on Storage Methods by Quantitative Scale

4. Survey on Land Transport of Ammonia Fuel

- 1) Survey on Transport Methods (Tanker, Trucks, Pipelines, etc.)

5. Survey on Ammonia Hazards and Safety Measures

- 1) Survey on Ammonia Hazards
- 2) Survey on Safety Measures for Ammonia

6. Cost Calculation for each Survey Item

- 1) Cost Estimates for Procurement, Maritime Transport, Storage, Land Transport & Safety Measures etc.



Delivery route of Ammonia to Malaysia (Estimated)

India Odisha → Malaysia

- IHI to develop Green Ammonia Project in Odisha, India with ACME
- Phase 1 (& Phase2 -Planning)
: 400,000 TPA each Green Ammonia
- Delivery date : Around 7-10 days



India Tamir Nadu → Malaysia

- Gentari to develop Green Ammonia PJ Titan Project
- Phase 1 : 500,000 TPA
- Phase 2 (Planning)
- Delivery date : Around 7-10 days



Singapore
(Ammonia Bunkering Hub)
→ → Malaysia
Delivery date: 4 - 5 days

Malaysia / Sarawak Area

Petronas Group
(PCG / Gentari) is planning /
producing Grey / Blue / Green
Ammonia to Malaysia Market .

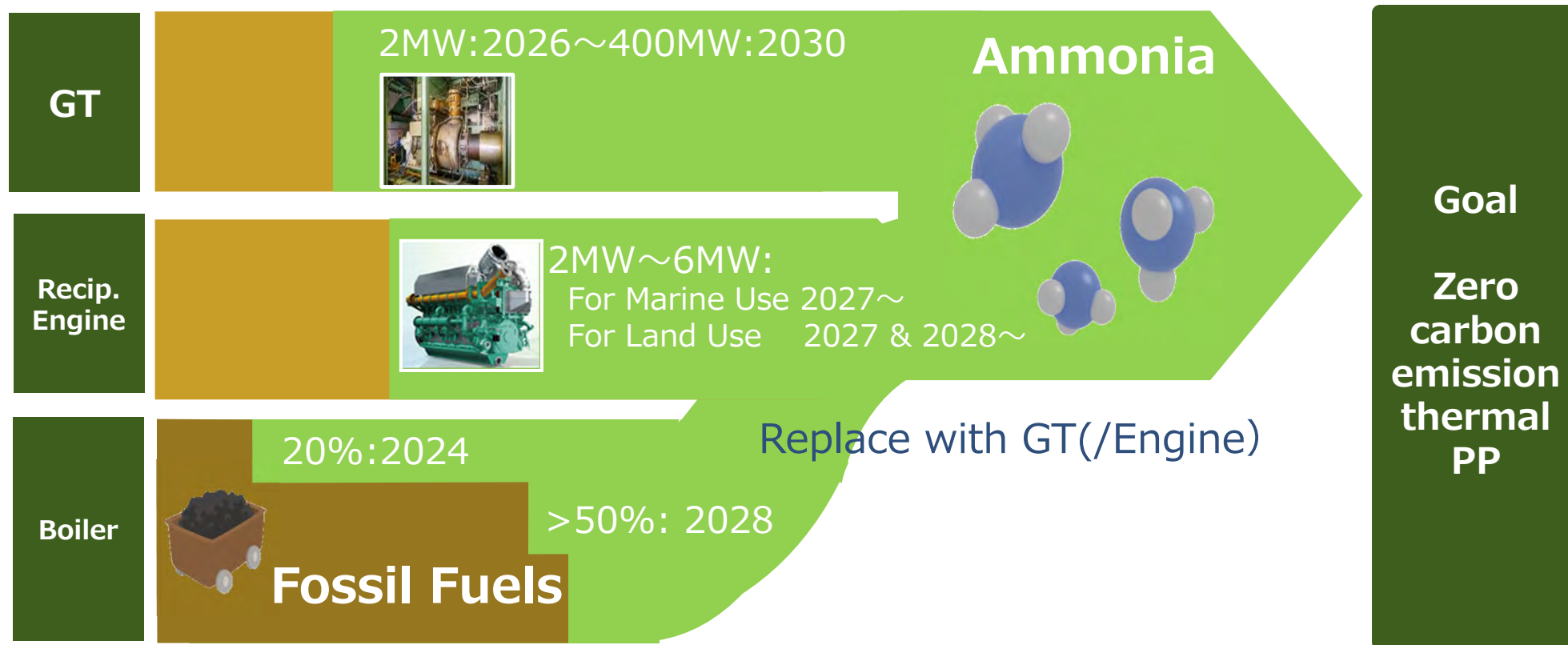
Delivery date: 4 - 5 days

IHI's Decarbonization Roadmap in Power Sector

Confidential

IHI

New-build & Retrofit of Ammonia combustion



Stepwise increase of co-combustion ratio toward zero CO₂ emissions



IHI's Ammonia Utilization Technology

IHI is developing a wide range of ammonia utilization technologies including the following:

Area	Technology/Application		Details
Power	Boilers		- Complete development of ammonia mono-firing burner (May 2022) - JERA/IHI conducted a 20% ammonia fuel conversion demonstration test at Hekinan Thermal Power Plant Unit 4 (1,000MW) (From Apr. to Jun. 2024)
	Large GTs		- IHI/GE signed a MOU regarding the development of large ammonia-fired GTs (Jan. 2023) - The goal is to commercialize GE's 6F.03, 7F and 9F GTs in 2030. 100% Ammonia Combustion Large Scale Test has been started and achieved the target milestone . (March., 2026)
Industry	Small GTs		- IHI's 2MW-class GT (IM270) successfully achieved the world's first 100% liquid ammonia combustion (Jun. 2022) - Long-term durability tests have been completed at IHI Aioi Works by end of March 2026.
	Industrial Furnaces		Successfully demonstrated over 20% ammonia fuel conversion at the naphtha cracking furnace at Idemitsu Tokuyama Plant (Feb. 2024)
	Engines		- Ammonia tug-boat demonstration test was conducted in Yokohama Port/ Tokyo Bay (From Jul. to Aug. 2024) - Ammonia/Diesel Dual Fuel engine(28ADF) for Land-use to be commercialized in 2027, & 100% Ammonia combustion engine (AGX) for Land-Use to be commercialized in 2028-2029.
Maritime			

Fuel Ammonia Utilization in Coal-Fired Power

IHI and JERA Complete Fuel Ammonia Substitution Demonstration Testing at Hekinan Thermal Power Station

-June 26, 2024-

Press Release

IHI announces the completion of pioneering fuel ammonia substitution demonstration testing at Unit 4 of the Hekinan Thermal Power Station of JERA. IHI and JERA conducted the world's first large-volume ammonia substitution demonstration testing (representing 20% of heating value) at a large-scale commercial coal-fired facility.

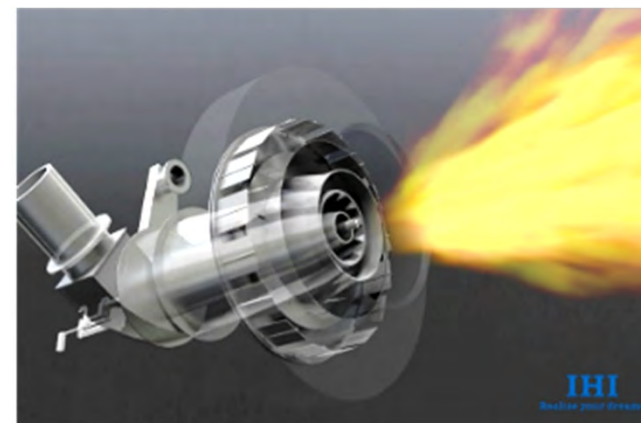
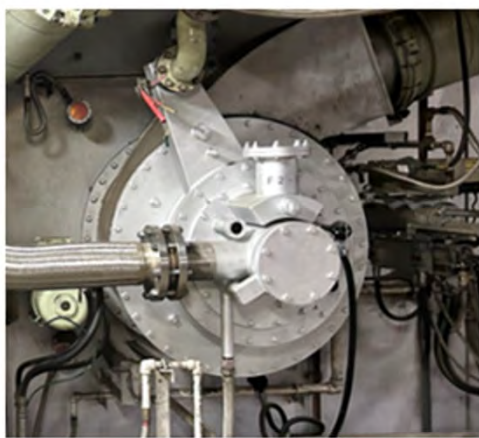
In this demonstration test, 20% substitution of fuel ammonia for operation of rated output of 1GW has been achieved successfully on April 10, 2024). This effort yielded favorable environmental outcomes. It is confirmed that carbon dioxide emissions at the unit fell around 20%, nitrogen oxide (NOx) emissions were equal to or less than when mono-firing coal before ammonia substitution, and sulfur oxide (SOx) emissions were down about 20%. Emissions of the potential greenhouse gas nitrous oxide (N₂O) were undetectable. IHI and JERA also confirmed that operability was equivalent to that before the conversion to fuel ammonia.

[URL] https://www.ihico.jp/en/all_news/2024/resources_energy_environment/1200954_13691.html

Jera

IHI

Realize your dreams



Demonstration Testing Facilities
(Left: Test burner / Right: Fuel ammonia tank)

NEDO : JPN16002

Fuel Ammonia Utilization in Coal-Fired Power

March 26 , 2026

Press Release

Construction of Fuel Ammonia Facilities for JERA's Hekinan Thermal Power Station Progressing

IHI Group company, IHI Plant Services Corporation have received an EPC (Engineering, Procurement and Construction) contract from JERA Co., Inc. (hereinafter "JERA") for four fuel ammonia storage tanks (total capacity: 160,000 tons) and associated facilities, to enable commencement of commercial operation with large-volume ammonia combustion (20% of heating value) at the Hekinan Thermal Power Station. Construction work is currently in progress.

Under this project, IHI Plant Services Corporation is responsible for the overall design of the ammonia tanks and associated facilities, the procurement of equipment, electrical and instrumentation components, piping, and other materials, as well as the construction work for the ammonia storage tanks.

【URL】 https://www.ihi.co.jp/en/all_news/2025/resources_energy_environment/1201927_13737.html



Fuel Ammonia Utilization in Coal-Fired Power

April 14, 2026

Press Release

IHI, Subsidiaries of Babcock Power Inc. Sign Strategic Collaboration Agreement for Low-Carbon Fuel Conversion Retrofits for Power Plant Boilers



IHI Corporation has entered into a strategic collaboration agreement with Babcock Power Services Inc. and Riley Power Inc. (collectively, BPS), leading U.S. companies with extensive experience in retrofit projects for power plant boilers. BPS will work together with IHI to accelerate the conversion of power plant boilers to low-carbon fuels.

This agreement establishes a framework for cooperation between IHI and BPS in the execution of fuel conversion projects for boilers. By leveraging the technological strengths, expertise, and know-how of these companies, we will jointly propose fuel conversion and mix-firing solutions to power plant operators worldwide, with a focus on low-carbon fuels such as ammonia.

BPS has successfully completed over 17 gigawatts of fuel conversions at power plants globally on a wide variety of boilers. BPS's proprietary technology solutions provide power plants with fuel flexibility, cost savings, and lower carbon solutions while achieving emissions and overall performance requirements.

Through the provision of decarbonization technologies for thermal power generation and models for effective utilization of ammonia fuel, IHI is committed to advancing a sustainable energy transition and contributing to the reduction of global environmental impact.

【URL】 https://www.ihi.co.jp/en/all_news/2026/resources_energy_environment/1201971_13814.html

Fuel Ammonia Utilization in Coal-Fired Power

June 29 , 2026

Press Release

Technical Verifications for Practical Application Completed on Ammonia Single Firing Burner for Thermal Power Plant Boilers

IHI announces it has developed an ammonia single firing burner for thermal power plant boilers and completed technical verifications for practical application.

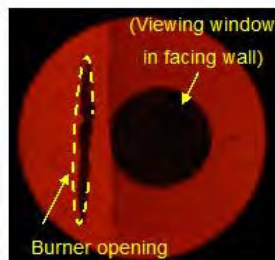
As a method to achieve even greater decarbonization, IHI has also worked on developing a burner capable of ammonia 60% high ratio firing and an ammonia single firing burner that can be implemented with minimal modifications on existing facilities.

The ammonia single firing burner was developed with the aim to increase the firing ratio of ammonia in stages until 100% ammonia firing can be realized, while minimizing modifications to existing boilers. In 2022, IHI successfully conducted stable ammonia single firing in a small combustion test facility. Because ammonia has the characteristic of low combustibility, IHI has repeatedly conducted testing at a large combustion test facility (image 1, image 2) at IHI's Aioi Works (Aioi City, Hyogo Prefecture) to realize stable combustion, reduce emissions of NOx and unburnt ammonia, and support load fluctuation which is required in thermal power plants.

【URL】 https://www.ihl.co.jp/en/all_news/2026/resources_energy_environment/1202094_13814.html



Image 1 Large combustion test facility at Headquarters Representative's Office, Aioi (Capacity: 10MW [by thermal input])



Shot with regular camera (flame not visible)



Shot with special camera (flame visible)

Image 2 Ammonia single firing flame

Gas Turbine using 100% Liquid Ammonia

株式会社IHI原動機
IHI Power Systems Co., Ltd.

CO₂-free power generation achieved with the world's first gas turbine using 100% liquid ammonia –Reduction of over 99% greenhouse gases during combustion–

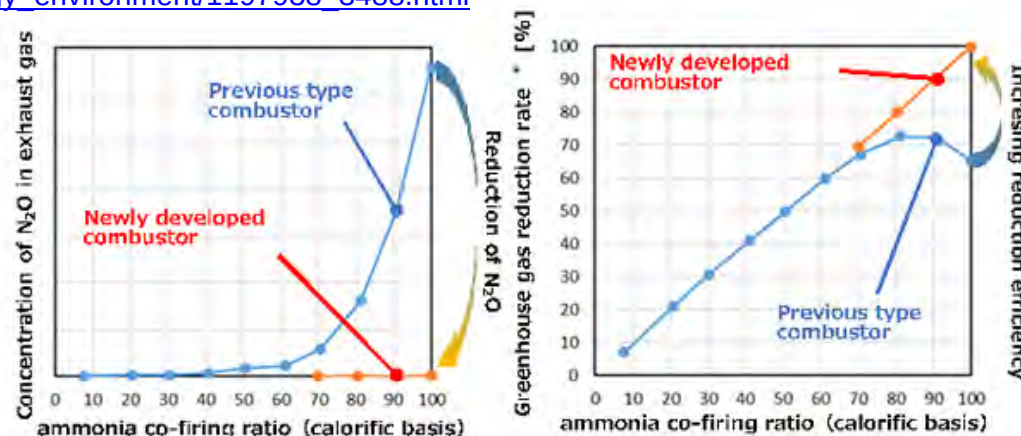
-June 16, 2022-

Press Release

IHI has succeeded in reducing greenhouse gases by over 99% during combustion of liquid ammonia in a 2,000-kilowatt-class gas turbine achieving truly CO₂-free power generation. Ammonia (NH₃) can be used in existing power generation facilities as a fuel which does not emit CO₂ when combusted. The combustion method which IHI is pioneering involves liquid ammonia being directly sprayed into the gas turbine combustor, which presents numerous advantages for social implementation, such as the simplification of the liquid ammonia fuel supply system from the storage tank to turbine, as well as improved controllability.

Until now, when operating gas turbines at ammonia co-firing rate of over 70%, nitrous oxide (N₂O) which has a greenhouse warming effect around 300 times that of CO₂, was susceptible to formation, nullifying the effect of reducing CO₂ emissions. As a result of mounting and testing a newly developed combustor on the 2,000-kilowatt-class gas turbine at IHI Yokohama Works, we were able to achieve a greenhouse gas reduction rate exceeding 99%, even when the ammonia fuel ratio is at 70~100%, and we verified the output of 2,000kW when mono-firing liquid ammonia. Looking forward, we will further reduce NO_x levels, improve operability, evaluate long-term durability, and proceed with efforts toward the practical application of a 100% liquid ammonia combustion gas turbine in 2025.

【URL】 https://www.ihico.jp/en/all_news/2022/resources_energy_environment/1197938_3488.html



Demonstration of Ammonia-powered Gas Turbine

IHI signs JCDA with PETRONAS and Gentari for demonstration of an ammonia-powered gas turbine

-April 10, 2026-

Press Release

IHI Corporation (IHI), a Japanese engineering company, has signed a Joint Collaboration Demonstration Agreement (the “Project”) with clean energy solutions company, Gentari through Gentari Hydrogen Sdn Bhd and subsidiaries of the PETRONAS Chemicals Group Berhad (PCG) group of companies, to carry out a demonstration of an ammonia-powered gas turbine (IM270 Gas Turbine) running fully on ammonia fuel. Under this Project, the parties will work together to deploy IHI’s fully ammonia-powered gas turbine at PCG’s ammonia plant at the PETRONAS Kertih Integrated Petrochemical Complex in Terengganu, Malaysia, using ammonia produced by the plant.

Once in operation, this will be a significant milestone in demonstrating a new end-use case for ammonia (as fuel) using the ammonia-powered gas turbine technology to produce clean power, thus showcasing its technical potential.

This Project was registered as an MOU-based initiative at the 1st AZEC Leaders’ Meeting and was subsequently selected for support under the Global South Future-Oriented Co-Creation Project (Large-Scale Demonstration in ASEAN Member Countries), an initiative administered by Japan’s Ministry of Economy, Trade and Industry (METI).

The parties are targeting the start of the demonstration in FY2027. (*FY2027 = based on the Japanese fiscal year). This demonstration could potentially be the world’s first fully ammonia-powered gas turbine to be deployed in a commercial plant.

IHI will continue to work closely with Gentari and the PETRONAS Group to advance decarbonization in the Asia-Pacific region and to expand the clean ammonia value chain in the world.



Signing of the JCDA ceremony photo session held on 26 March at METI in Japan, attended by PETRONAS (Tan Sri Tengku Muhammad Taufik, PETRONAS President & Group CEO and Gentari Chairman) and IHI (Hiroshi Ide, IHI Corporation President & CEO)



Development of large-scale ammonia gas turbine system **IHI**

Jan 24, 2024

Press Release

GE Vernova and IHI move to the next phase of the technology roadmap aiming to develop a 100% ammonia capable gas turbine combustion system by 2030

9F.04 : Source : GE

Under a Joint Development Agreement (JDA), GE Vernova and IHI will collaborate to develop a new gas turbine combustor capable to use ammonia for power generation compatible with GE Vernova's 6F.03, 7F and 9F gas turbines.

In this engineering phase, combustion technology concepts will be evaluated for their ability to meet key operational requirements and the impact on the entire power plant. GE Vernova and IHI plan to develop a two-stage combustor configured to burn up to 100% ammonia to comply with emission requirements. Combustion testing for the maturation of the new combustor design will take place in IHI's facilities in Japan.

In addition to installation for new builds, the new combustor is expected to offer an alternative path to decarbonize existing gas turbine power plants. Without replacing the existing gas turbine, power plants operating on conventional fuels can be upgraded to operate on fuel that does not produce carbon emissions when combusted by simply changing the natural gas combustor to a new ammonia combustor with minimum upgrades to the fuel system and associated balance of plant systems.

【URL】 https://www.ihi.co.jp/en/all_news/2023/resources_energy_environment/1200574_3523.html

Development of large-scale ammonia gas turbine system **IHI**

March 18, 2026

Press Release

IHI and GE Vernova achieve milestone with 100% Ammonia Combustion in Large Scale Test



IHI and GE Vernova have successfully demonstrated combustion of 100% ammonia using full-scale components at pressures, temperatures, and flows matching full-load conditions for GE Vernova's F-Class gas turbines.

Emission levels are aligned with the companies' development roadmap; IHI and GE Vernova are continuing testing with prototype combustors.

This project is expected to play a meaningful role in decarbonizing the energy sector by enabling reduced or zero CO₂ emissions during combustion.

【URL】 https://www.ihi.co.jp/en/all_news/2025/resources_energy_environment/1201919_13737.html

Utilization of Ammonia in the Chemical Industry

Apr 26, 2024

Press Release

Idemitsu Kosan Demonstrates IHI Group's Ammonia Combustion Technology in Naphtha Cracking Furnace at Tokuyama Complex in Move toward Carbon Neutrality for Chemical Industry



出光興産徳山事業所ナフサ分解炉（提供：出光興産）



アンモニア燃焼時のナフサ分解炉内の様子



基礎燃焼試験炉（IHI相生事業所内）

IHI Corporation and subsidiary IHI Plant Services Corporation today announced that they have provided ammonia single-fuel burners and installed facilities for an ammonia combustion demonstration that Idemitsu Kosan Co., Ltd., undertook at a commercial naphtha cracking furnace at its Tokuyama Complex in Shunan City, Yamaguchi Prefecture, from February 6 through 8. Such furnaces operate at high temperatures to crack naphtha and produce ethylene and propylene that serve as basic raw materials for petrochemical products. This world-class technological achievement is the fruit of IHI's combustion technology and Idemitsu's operational expertise.

IHI began developing an ammonia burner for naphtha cracking furnaces in 2021. It assessed the combustion performances of several prototype burners at a one-megawatt thermal basic combustion test furnace at the Aioi Works in Hyogo Prefecture. The company chose the best design and confirmed that it met all performance requirements, including emissions limits for nitrogen oxide and unburned ammonia.

Testing confirmed that there were no environmental performance issues. Combustion adjustments reduced nitrogen oxide emissions from exhaust gases to below permissible limits, with no detection of unburned ammonia. IHI aims to deploy its burner at other naphtha cracking furnaces to help decarbonize the economy by expanding the use of ammonia as a fuel. It also plans to leverage technical expertise from this demonstration to enhance other industrial furnaces and burners.

【URL】 https://www.ihi.co.jp/en/all_news/2024/resources_energy_environment/1200934_13691.html

Development of ammonia-fueled engines

Aug 23, 2024

Press Release

Ammonia-Fueled Vessel was Completed - World's First Commercial-Use Ammonia-Fueled -



アンモニア燃料タグボート「魁」

On August 23, the ammonia-fueled tugboat Sakigake was completed by NYK Line (NYK) and IHI Corporation consolidated subsidiary IHI Power Systems Co., Ltd. (IHI Power Systems) in cooperation with Nippon Kaiji Kyokai (ClassNK). This is the world's first ammonia-fueled vessel for commercial use. The NYK Group company Shin-Nippon Kaiyosha will employ the vessel in tugboat operations in Tokyo Bay over a three-month demonstration period.

This vessel was a Green Innovation Fund Project initiated in October 2021 under Japan's New Energy and Industrial Technology Development Organization (NEDO) to develop vessels equipped with domestically produced 280mm bore four-stroke ammonia-fueled engines made by IHI Power Systems.

As part of the development of vessels equipped with domestically produced ammonia-fueled engines through NEDO's Green Innovation Fund Project, NYK, IHI Power Systems, Japan Engine Corporation, Nihon Shipyard Co., Ltd., and ClassNK are conducting research and development on ammonia-fueled medium gas carrier (AFMGC). This vessel is scheduled for delivery in November 2026. IHI Power Systems will be made to develop a 250 mm bore four-stroke engine for the auxiliary engine of this vessel.

【URL】 https://www.ihl.co.jp/en/all_news/2024/resources_energy_environment/1201003_13691.html

Ammonia Fueled Recip. Engine 28ADF

IHI Ammonia Dual Fuel Engine Type: 6L28ADF (for Marine Use)



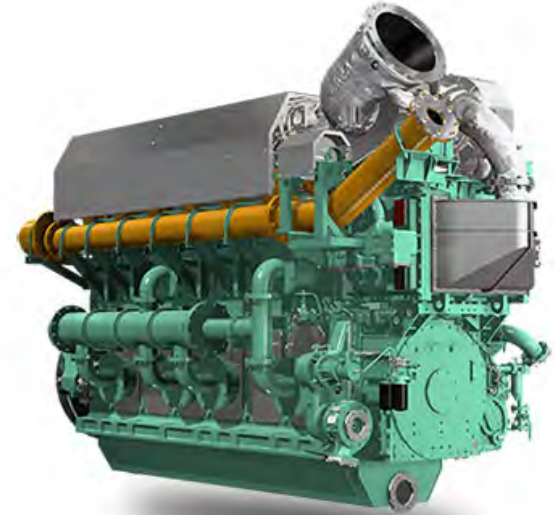
Ammonia Engine 28ADF

IHI Ammonia Firing Recip. Engine

Ammonia Dual Fuel Engine 6L28ADF

Embark on a new era of zero carbon with ammonia.

- ▶ Achieved a maximum co-firing rate of **95%** with ammonia.
- ▶ Achieved a reduction of over **90%** in GHG emissions.
(Already achieved the GHG reduction target for 2040 set by the IMO.)
- ▶ Making a significant contribution towards achieving carbon neutrality and **zero** GHG emissions by 2050.



6L28ADF

For Marine Use Engine, we have received the Class NK Type approval Certification & EIAPP certification on April, 2024.

For Land Use Engine,

- 1) 28ADF Dual Fuel Engine will be ready for commercialization from 2027.
(Ammonia Fuel / Diesel or Light Fuel Oil)**
- 2) 28AGS-AM Ammonia 100% Engine will be ready for commercialization from 2028.**

https://www.ihi.co.jp/ips/english/products_marine/niigata/ammonia_dual_fuel.html

Ammonia-powered Dual Fuel Recip. Engine

IHI Power Systems and IHI Corporation Commence Demonstration of Ammonia-Fueled Reciprocating Engine for Land-Based Power Generation

-July 15, 2026-

Press Release

IHI Power Systems Co., Ltd. (IPS), a member of the IHI Group, and IHI Corporation are pleased to announce the commencement of demonstration operations of a land-based power generation plant powered by a 6,000 kW-class ammonia-fueled reciprocating engine (18V28ADF Engine) at its Ota Works in Gunma Prefecture, Japan.

Leveraging technologies developed through its pioneering work on ammonia-fueled marine engines, IPS is advancing the development of a land-based ammonia-fueled reciprocating engine capable of achieving an ammonia fuel ratio and greenhouse gas (GHG) emissions reduction rate of more than 90%.

Through this demonstration program, IPS will verify the safety and operability of the complete power generation system, including auxiliary facilities. The demonstration testing is scheduled for completion during Japanese fiscal year 2026, with commercial sales planned to commence in Japanese fiscal year 2027.



IHI Power Systems Co., Ltd.

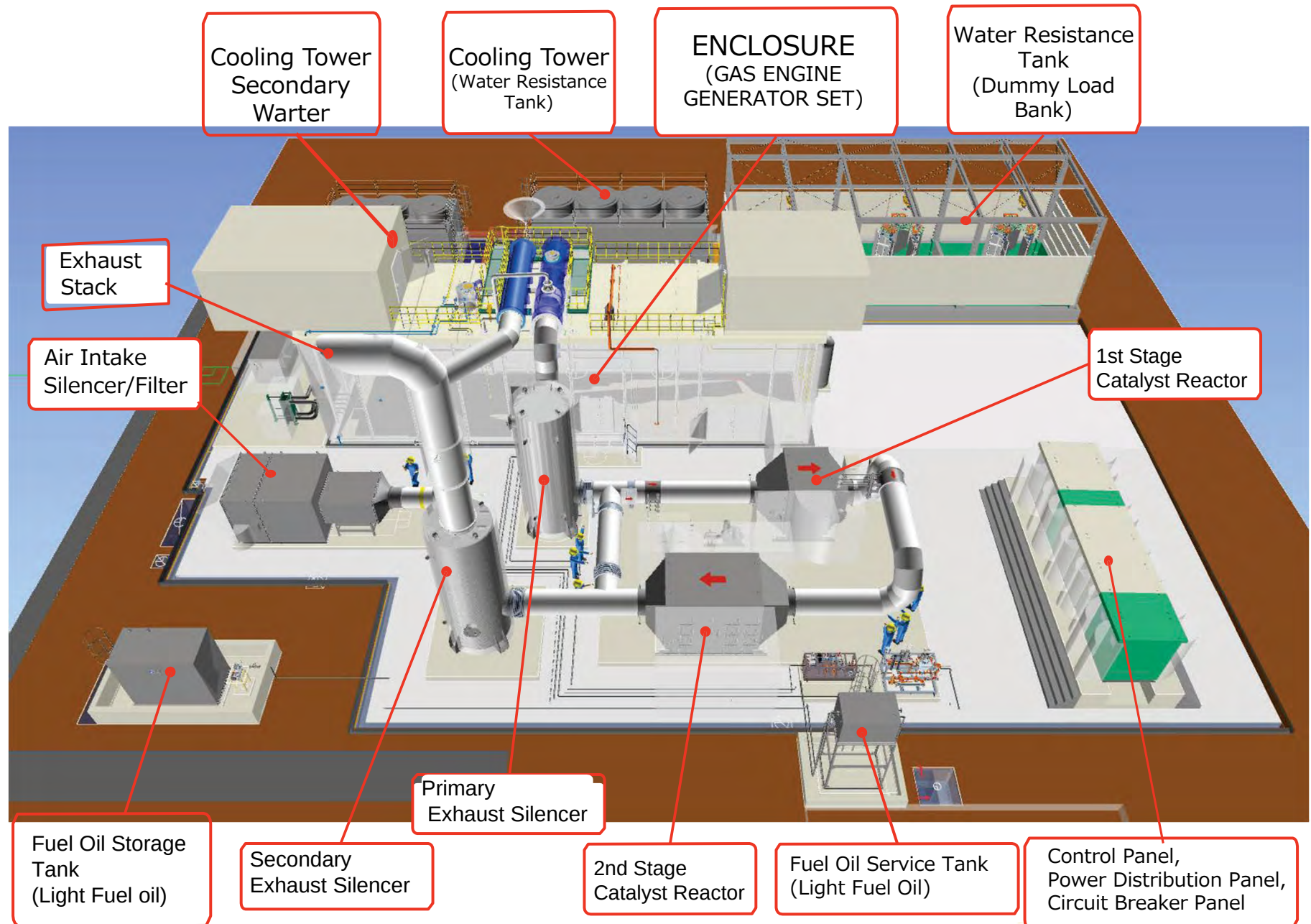
N/GATA

【URL】 https://www.ihi.co.jp/en/all_news/2026/resources_energy_environment/1202137_13814.html

28ADF Dual Fuel Engine Demonstration Plant in IHI Power Systems Ohta Works (Layout) - 6MW

Confidential

IHI



IHI Power Systems Co., Ltd. **N/GATA**

Total Green Solution & Management System, targeting NET ZERO

IHI Scope of Work - One Stop Solution by IHI Group

Replacement of existing
Diesel Engine
/Gas Engine



Green
Electricity



Household



Hotels,
Resorts & Restaurant



Industry

Green Steam/Heat

Green District Cooling
&
Green Chilled Water

Additional System



Solar PV
Power Plant



Battery
Energy
Storage
System
(BESS)



Water
Electrolyser



Fuel Cell



IHI Ammonia Combustion Engine
+ HRSG
Cogeneration System

IHI Energy
Management
System (EMS)

IHI Ammonia Combustion Engine – Performance

<28ADF Series> (For Land Use) <Target>

Type	6L28ADF	8L28ADF	9L28ADF	12V28ADF	16V28ADF	18V28ADF
Bore x Stroke	280 mm x 390 mm					
Engine speed (min-1)	750(50Hz) / 720(60Hz)					
Engine output (kWe) 50Hz/60Hz	2000 / 1900	2500 / 2400	3000 / 2850	4000 / 3800	5000 / 4800	6000 / 5700
Generator Efficiency (%)	96.0	96.0	96.0	96.5	97.0	97.0
BMEP (bar) 50Hz/60Hz	23.1 / 22.9	21.7 / 21.7	23.1 / 22.9	23.0 / 22.8	21.5 / 21.5	22.9 / 22.7
Starting system	Air motor start + Diesel start					
Type of fuel	MDO / ammonia					

***Conditions :**

(1) Under ISO Condition (Ambient Temp: 25°C, Cooling Water Temp at A/C inlet :25°C, Atmospheric Pressure: 101.3 kPa (760mmHg))

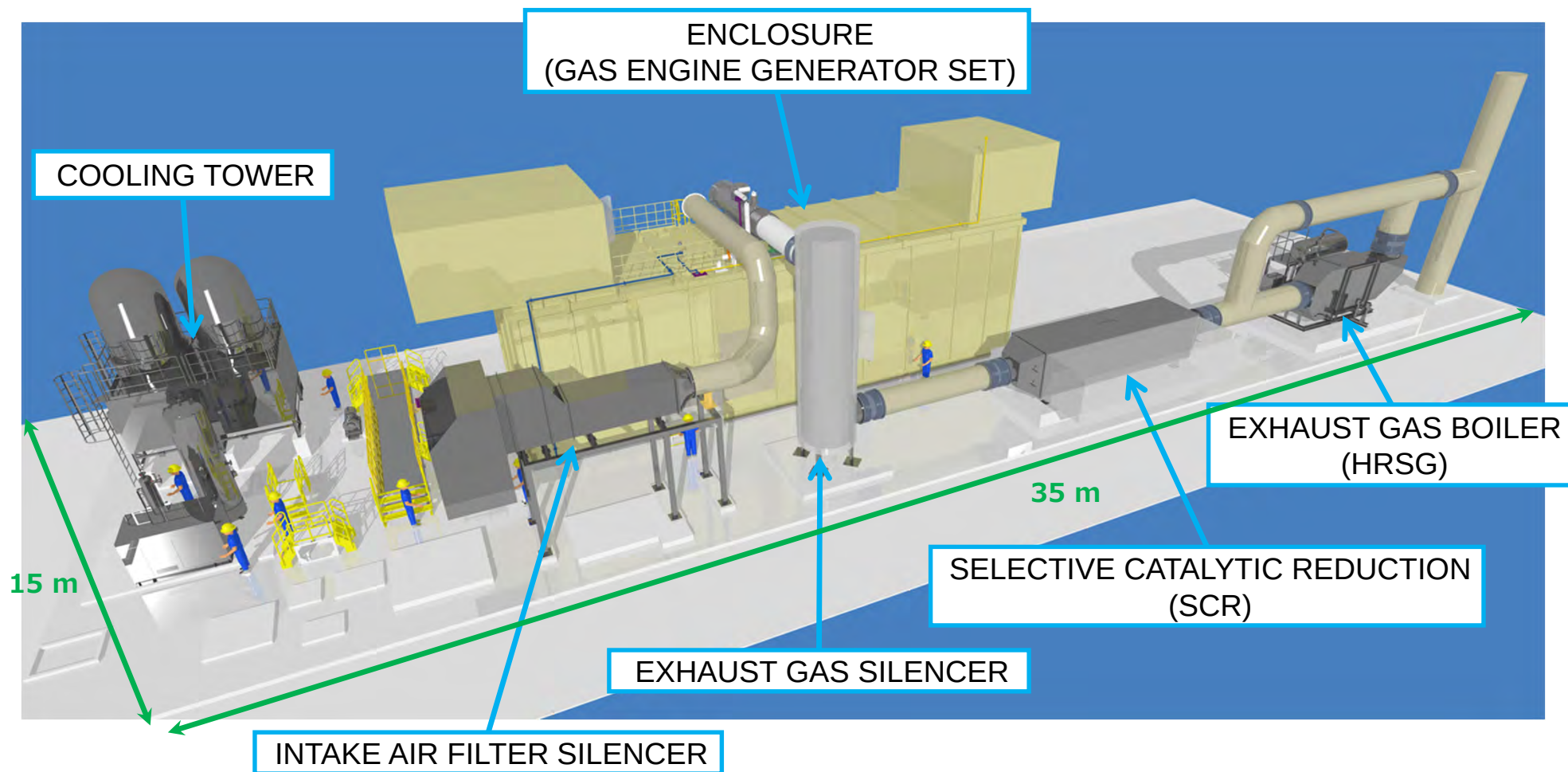
(2) Efficiency is subject to ISO3046 reference with no tolerance at rated output and calculated based on fuel having a net calorific value of 42.7 MJ/kg (10,200 kcal/kg); net consumption calculated excluding leakage and treated losses.

Ammonia Gas Turbine IM270 & Recip. Engine 28ADF

IHI Ammonia Combustion Gas Turbine / Recip. Engine

Item	Ammonia Combustion IM270 Gas Turbine	Ammonia Combustion Recip. Engine 28ADF Engine
GT/Engine Output 【kW】	2,000kW	2,000kW
Steam Supply Amount 【Ton/h】 (One Unit)	Approx. 5.0Ton/h	Approx. 0.8Ton/h (Estimated / GT Basis)
Ammonia Fuel Consumption 【Ton/h】 (One Unit)	Approx. 1.6Ton/h	Approx. 0.9Ton/h (Estimated /GT Basis)
Installation Unit (s)	1~4 Unit(s)	1~4 Unit(s)
GT/Engine Output 【kW】	2,000~8,000kW	2,000~8,000kW
Steam Supply Amount 【Ton/h】	Approx. 5.0~20.0Ton/h	Approx. 0.8~3.2Ton/h
Ready for Commercialization	From 2026	From 2027

IHI Ammonia Combustion Engine Plant Layout Image (For Land Use) 6L28ADF x 1 Unit Case (2MW)

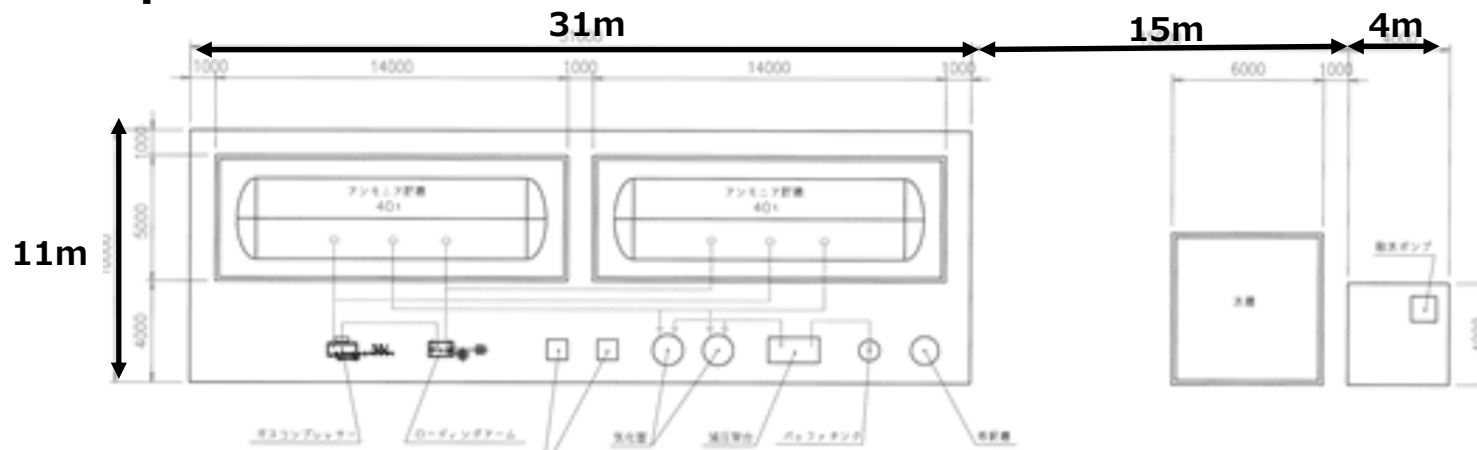


IHI Ammonia Combustion Engine Plant Layout Image

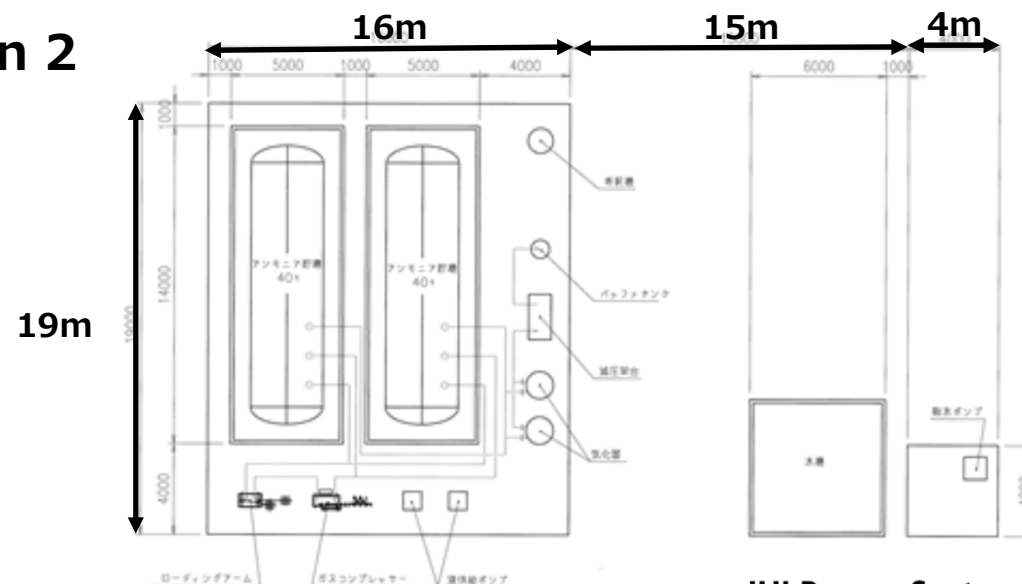
(For Land Use) Ammonia Fuel Storage & Supply System

100kL Tank x 2 Units (40 ton/unit)

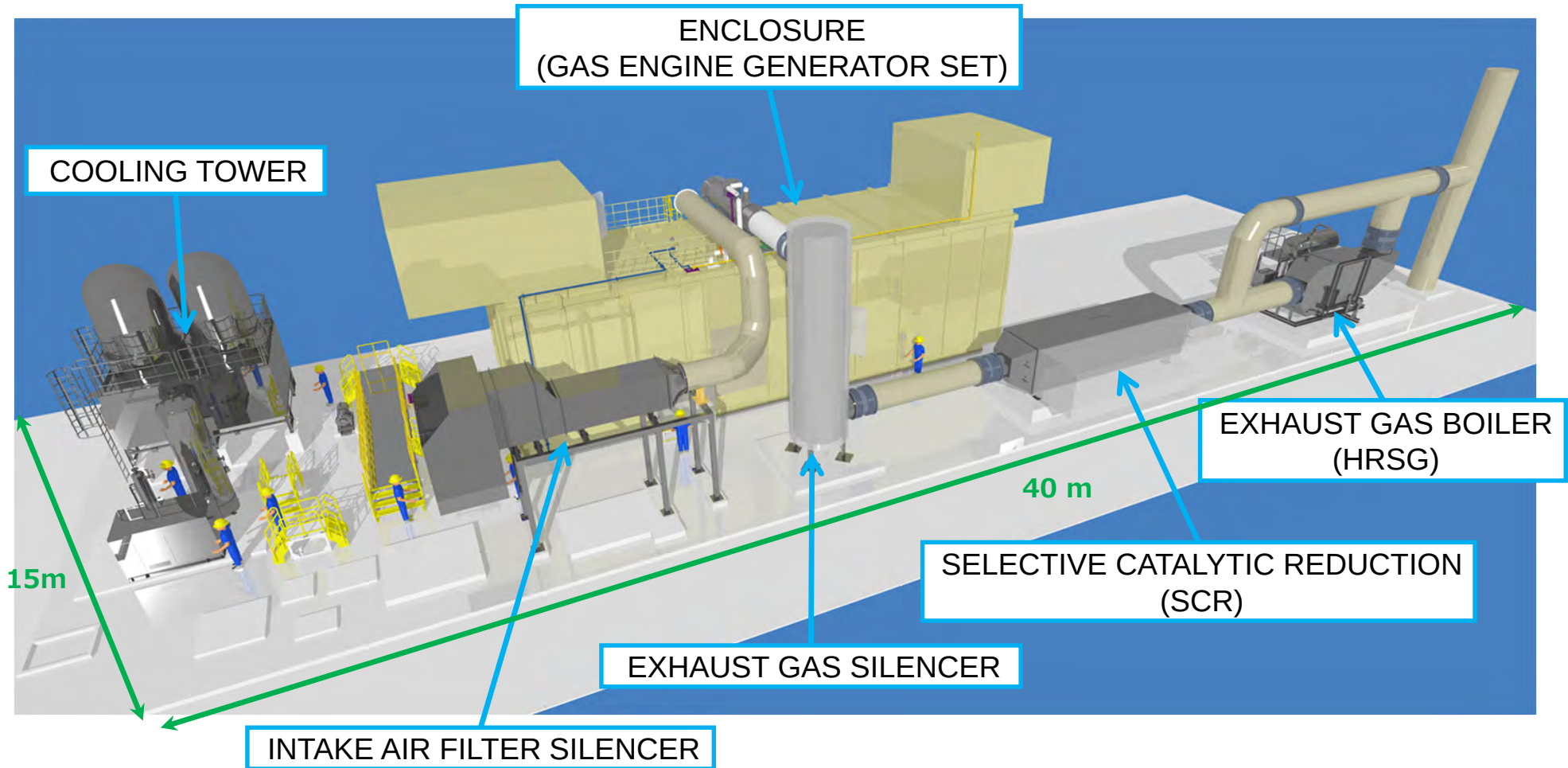
Layout : Option 1



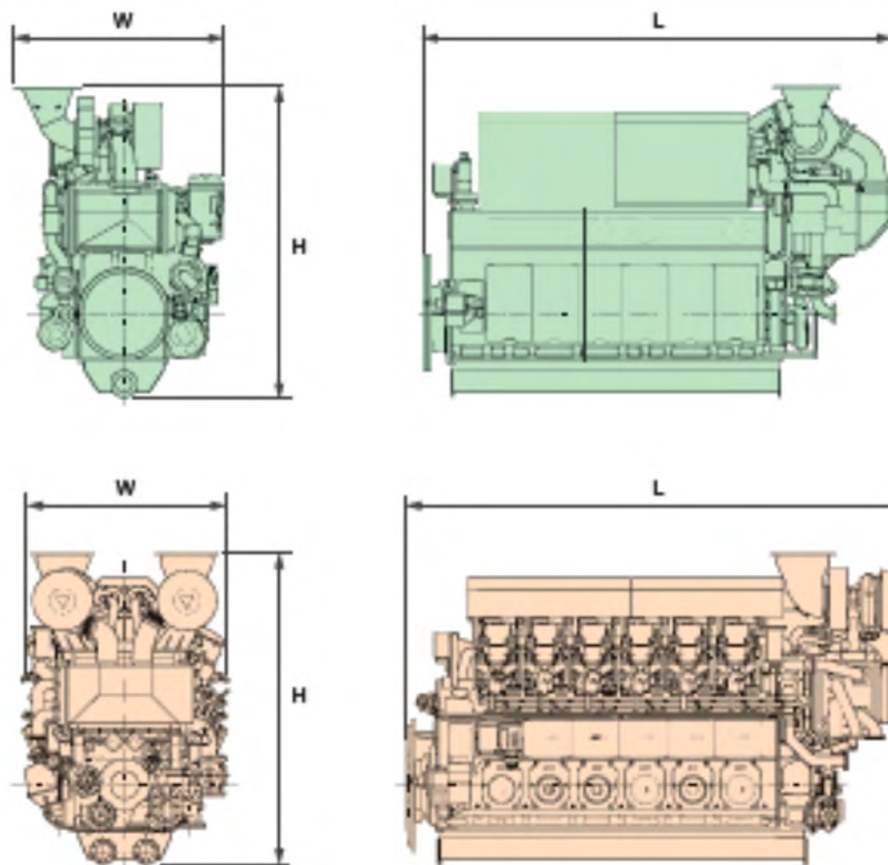
Layout : Option 2



IHI Ammonia Combustion Engine Plant Layout Image (For Land Use) 18V28ADF x 1 Unit Case (6MW)



IHI Ammonia Firing Recip. Engine Size (Image)



**These Pictures are just image, reference from 28AHX Engine (Diesel Engine).*

Engine Dimension (mm)

MODEL	L	W	H	Engine Dryweight (Ton)
6L28ADF	4450	2200	3525	24
8L28ADF	5350	2300	3525	30.5
9L28ADF	5800	2300	3525	33
12V28ADF	5350	2850	3370	38
16V28ADF	6500	2850	3370	47
18V28ADF	7000	2850	3370	52

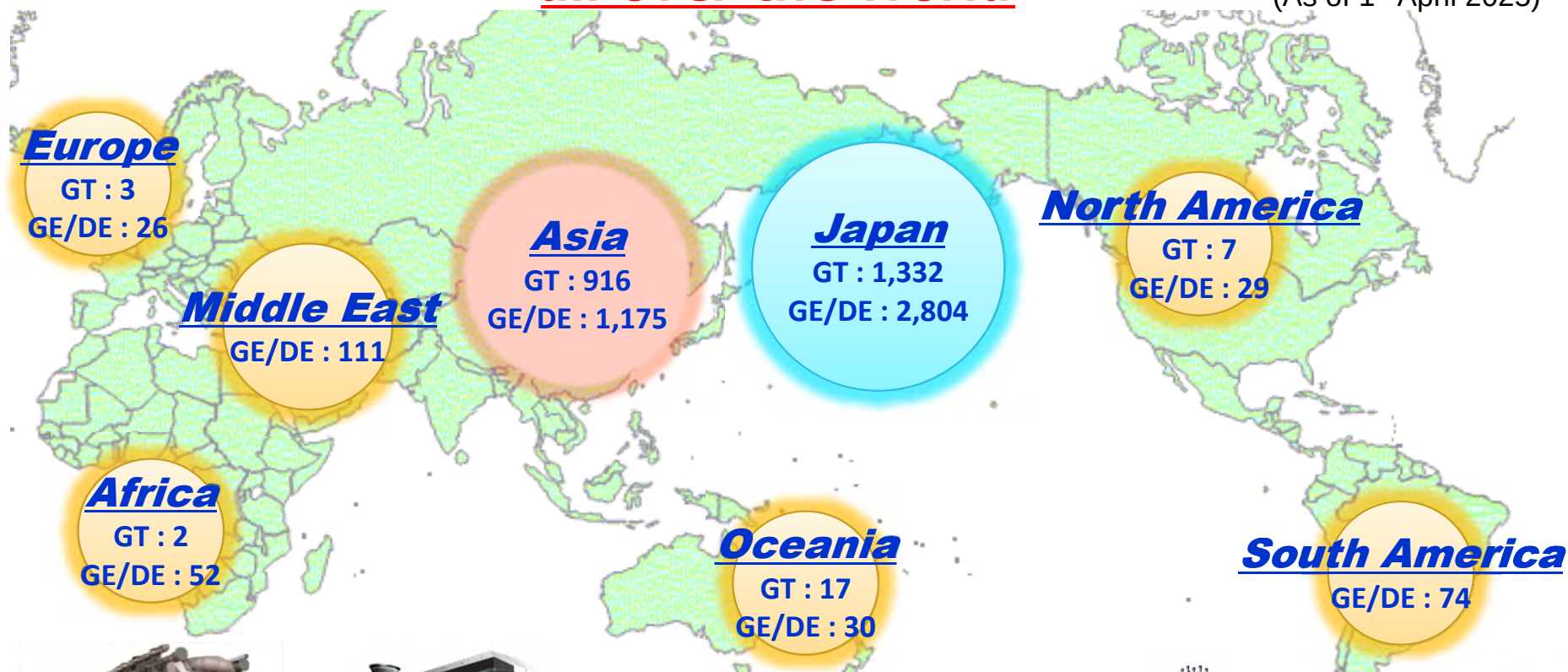
**Above Size & Dryweight are just for your reference size, reference from 28AHX Engine (Diesel Engine).*

**Specification may change without notice.*

Delivery Record (Land Use Product)

IHI

Enrich Experience by Highly Reliable Engine & Turbine all over the World

(As of 1st April 2025)

28AGS



28AHX



LM6000



IM270

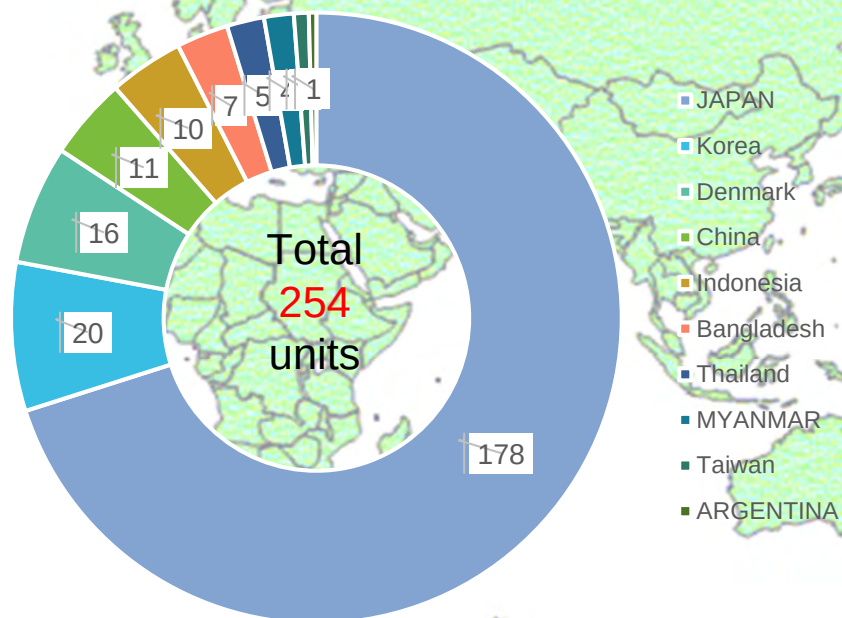


Taurus70

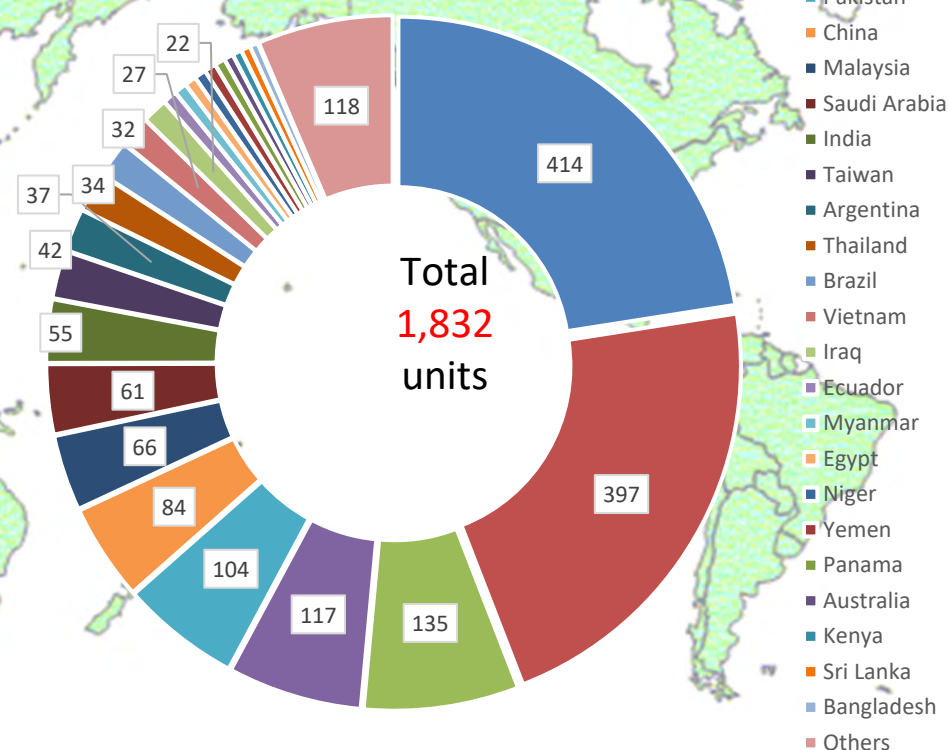
Recent Delivery Record for IPS (Niigata) Reciprocating Engine

As of June, 2024

Gas Engine Supply Record as of June 2024 (Unit)



Diesel Engine Supply Record as of June 2024 (Unit)



Network of IHI Power Systems (Production Bases and Branches)

IPS Niigata Gas Turbine Plant



NICO Precision Co., Inc.
(100% subsidiary)

IPS Niigata Engine Plant &
Niigata Foundry Plant



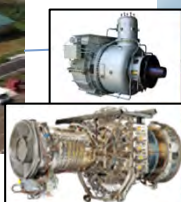
Head Office



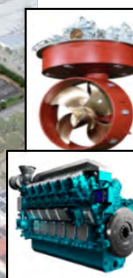
Mizuho Works (IHI Corporation)



Kure Works (IHI Corporation)



IPS Ohta Plant



IHI Power Systems HQ (JAPAN)

Overseas Offices

IHI Malaysia Base

Niigata Power Systems Philippines Inc.

IHI Power Systems Thailand Co, Ltd.

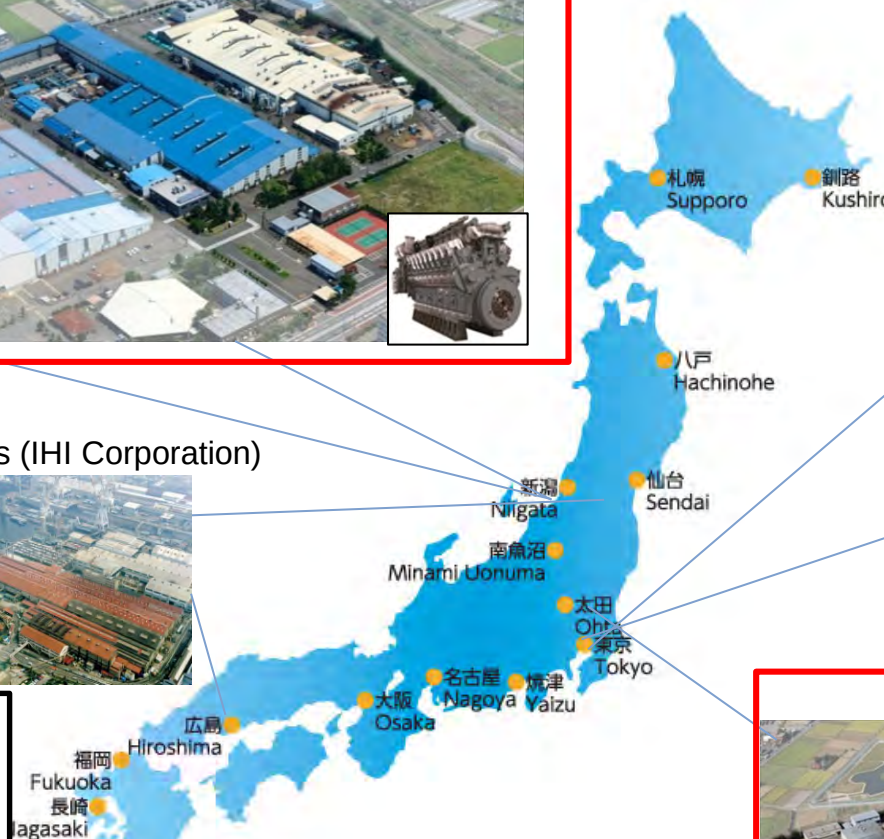
Niigata Power Systems (Singapore) Pte. Ltd.

IHI ASIA PACIFIC (Singapore)

Niigata Power Systems Co., Ltd. Seoul Office

Niigata Power Systems (Europe) B.V.

Niigata Power Systems (Shanghai) Co., Ltd.



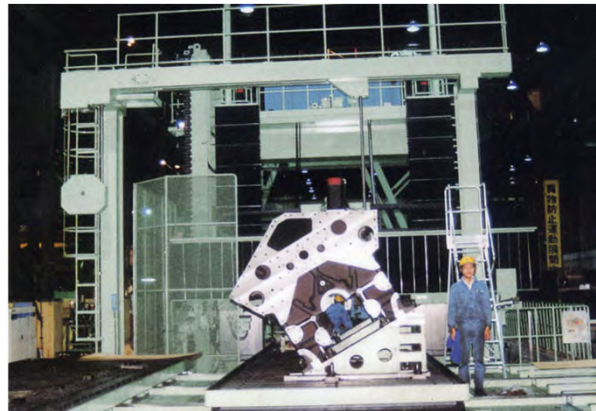
NIIGATA Engine Plant & Foundry Plant



- Location : Niigata, Japan
- Production : Diesel Engine (Medium Speed) , Gas Engine, Cast Iron for Engine Equipment etc.)



Casting



Machining (Five-face machining)



Shop Test



Casting products (Crank cases)



Assembling



Packing for shipping

Ohta Plant

Confidential

IHI



- Location : Gunma, Japan
- Production : Diesel Engine (High / Medium Speed) , Dual Fuel Engine, Z-Peller



Machining



Assembling



Z-PELLER® assembling



Assembling



Shop test



Z-PELLER® shop test

